Chemical Theory II
Statistical Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Recap of thermodynamics
  - Second law: $dS_{univ} > 0$ in a spontaneous process
  - (Ir)reversible processes, state & path functions, $dS = \delta q_{rev} / T$
  - First law: $\Delta U = q+w$
  - Gibbs energy $G = H - TS$ is a minimum at equilibrium at constant p and T.
- The Helmholtz energy
  - $A = U - TS$
  - A is a minimum at equilibrium at constant V and T.
- Thermodynamic master equations:
  - $dU = TdS - pdV$ (1st and 2nd law combined)
  - $H = U + PV$; $A = U - TS$; $G = U + PV - TS$
- Legendre transform

Course outline

- Statistical thermodynamics - key ideas

Course outline

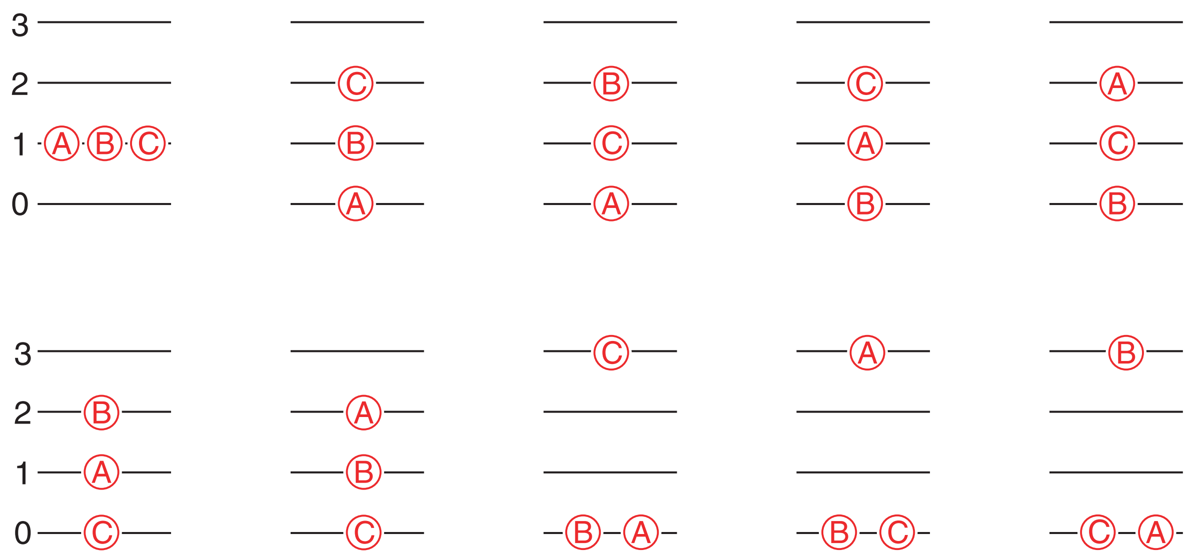
- In this section we will introduce the basic ideas and postulates that lie at the heart of statistical thermodynamics. At first they may seem quite abstract, but once we have established them we will quickly be able to move on to practical applications in chemistry.

Macrostates and microstates

- We start out by thinking about an isolated system. The thermodynamic state of such a system is specified completely if we know its internal energy (***U***), its volume (***V***) and the number of particles (***N***) present. All of the other thermodynamic parameters of the system can be found by using thermodynamic relationships. *The state of the system described by particular values of **U**, **V** and **N** is called a **macrostate***.
- However, knowing the macrostate tells us nothing about which particle is occupying which energy level. Remember that each molecule has available to it a large number of energy levels, and that in a macroscopic sample there are a very large number of molecules. The number of different ways that the molecules can be assigned to energy levels is staggeringly large; the thermodynamic state gives us no information about which of the many possible arrangements is adopted.
- *A particular arrangement of the particles in the energy levels (states) is called a **microstate***. For a macroscopic sample the number of possible microstates that correspond to a particular macrostate is very large. A possible microstate is one which has the correct values of ***N***, ***V*** and ***U***.

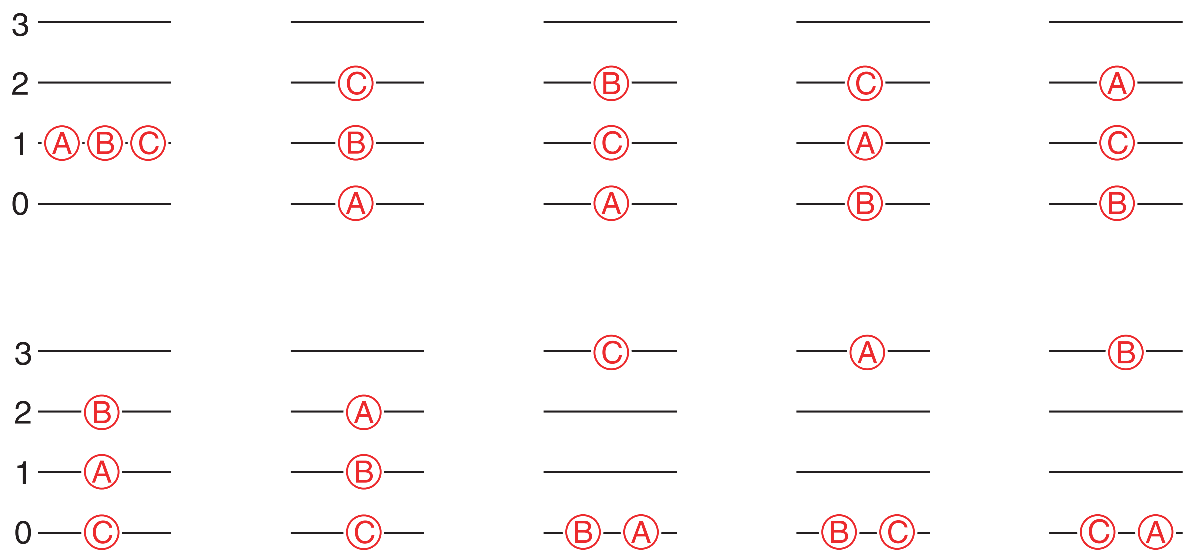
Counting microstates

- To grasp the idea of what different microstates are, let us consider a very simple system. We will have 3 distinguishable particles, A, B and C and these will have available to them a set of energy levels with energies 0, 1, 2, 3 . . . in arbitrary units. The total energy of the system will be fixed at 3 units. *As we have specified N, V and the total energy (U), the macrostate is defined*.
- There are 10 different ways in which the particles can be arranged amongst the energy levels such that the total energy is 3 units:



Counting microstates

- Each one of these is a *microstate*; note that some of the microstates only differ because the particles are *distinguishable*. There are 10 microstates which are consistent with the macrostate; the number of microstates, $\Omega = 10$.
- It is easy to convince ourselves that if t*he total energy increases, or the number of particles increases, the number of microstates will increase*. The number of microstates also increases as the separation of the energy levels decreases. In fact the number of microstates increases very rapidly with ***N*** and ***U***.



Example

- Consider the same set of energy levels (0, 1, 2..) as described in the text. How many microstates are there for the three particles if the energy is increased to four units?
- For the same set of energy levels, find the number of microstates for $E = 3$, $N = 4$
- Suppose that the spacing of the energy levels is halved. How many microstates are there for $N = 3$, $E = 3$?

Postulate of equal probabilities

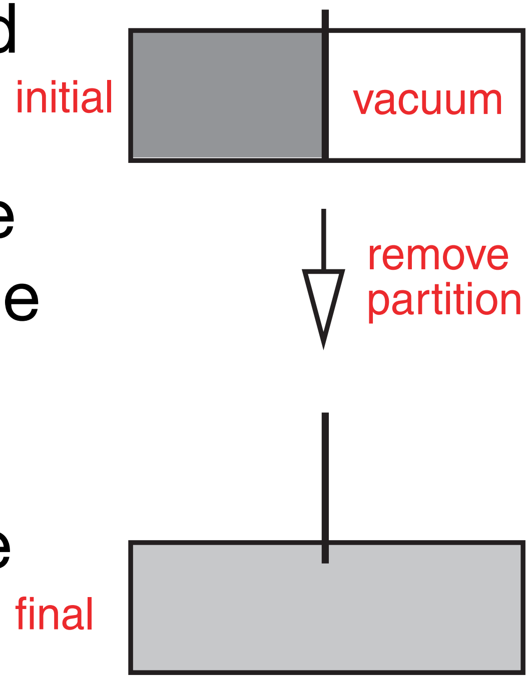
- We now assert that there is no reason for the system to prefer any one microstate over any of the others. We can imagine that the particles in the system constantly rearrange themselves amongst the levels thus moving from one microstate to another; after a sufficient time the system will have been in each of the microstates an equal number of times.
- Put more formally, this first postulate is:
  Postulate1: *for an isolated system, the probability that the system will be in any one accessible microstate is 1/Ω, where Ω is the number of microstates.*

Postulate of equal probabilities

- *The greater the number of microstates, the less probable it is that the system will be in any particular microstate* (loosely – the less ‘time’ the system will spend in any particular microstate). *By ‘accessible microstate’, it is meant a microstate which has the correct number of particles and total energy*.
- For the example given above the probability of finding the system in any one of the ten microstates is $1/\Omega = 0.1$; this means that the system spends one-tenth of the time in each of the microstates.

Irreversible processes

- Postulate 1 can be used to explain why some processes are spontaneous (or irreversible). Recall that in an isolated system, a spontaneous process must be accompanied by an increase in entropy.
- An example of a spontaneous process is the expansion of an ideal gas into a vacuum. The situation is shown opposite: a partition constrains an ideal gas to half of the vessel, the other half is evacuated. On removing the partition the gas expands to fill the entire vessel. As the gas is ideal, there is no change in energy. Clearly the process is spontaneous and thermodynamics tells us that the entropy change is $\boldsymbol{R \ln (V_{final} / V_{initial})}$ per mole.



Irreversible processes

- For an ideal gas, the energy levels available to the particles are those due to translation; to a good approximation we can model these using the energy levels of a particle in a box. In one dimension these are

$$E_n = \frac{n^2 h^2}{8ma^2} \qquad n = 1,\, 2,\, 3\ \ldots$$

where ***a*** is the length of the box. In three dimensions the energy levels are just the sum of three such terms, one each for x, y and z. *The key point is that the energy levels get closer together as the size of the box increases*.

Irreversible processes

- So, in the case of an expansion, the energy levels are closer together in the final state than they were in the initial state. We saw previously that the number of microstates goes up as the energy levels get closer together so it follows that
  $\boldsymbol{\Omega_{final} > \Omega_{initial}}$
  where $\boldsymbol{\Omega_{initial}}$ and $\boldsymbol{\Omega_{final}}$ are the number of microstates of the initial and final states, respectively.

Irreversible processes

- As each microstate is equally probable, it follows that the system will spend more time in microstates belonging to the final state than in microstates belonging to the initial state, simply as there are more of the former. *If Ω_{final} >>> $\Omega_{initial}$ the system will effectively only ever be found in microstates belonging to the final state, and so it will appear that the process is irreversible.* In practice, for any finite spontaneous change it is always the case that **Ω_{final} >>> $\Omega_{initial}$.**
- Generalising this, we can say that, as a consequence of Postulate 1, *a system evolves into a state with the maximum number of accessible microstates. The existence of irreversible processes are thus explicable in terms of the idea of equally probable microstates.*

Boltzmann's definition of entropy

- The discussion in the previous section gives a strong hint that there must be a connection between entropy and the number of microstates, $\boldsymbol{\Omega}$. We follow Boltzmann and state the second postulate
- ***Postulate 2:***
  The entropy of an isolated system is given by $S = k \ln \Omega$, where k is a universal constant.
- We can rationalise the form of this relationship by thinking about two systems, A and B, and what will happen when we bring them together to form a single system. The entropies of system A and B are $\boldsymbol{S_A}$ and $\boldsymbol{S_B}$ respectively; the number of microstates of A and B are $\boldsymbol{\Omega_A}$ and $\boldsymbol{\Omega_B}$ respectively.

Boltzmann’s definition of entropy

- When we bring the two systems together we expect the entropies to add: $\boldsymbol{S_{tot} = S_A + S_B}$. However, we expect the number of microstates available to the combined system to be equal to the product of the number available to each separately: $\boldsymbol{\Omega_{tot} = \Omega_A \Omega_B}$. This is because each microstate of A can go with all of the microstates of B and vice versa.
- Using Boltzmann’s definition of entropy we have the following

$$S_{\mathrm{A}} = k \ln \Omega_{\mathrm{A}} \quad S_{\mathrm{B}} = k \ln \Omega_{\mathrm{B}} \quad S_{\mathrm{tot}} = k \ln \left(\Omega_{\mathrm{A}} \Omega_{\mathrm{B}}\right)$$

which are consistent with the expected additivity of $\boldsymbol{S_A}$ and $\boldsymbol{S_B}$

$$\begin{aligned} S_{\mathrm{tot}} &= S_{\mathrm{A}} + S_{\mathrm{B}} \\ &= k \ln \Omega_{\mathrm{A}} + k \ln \Omega_{\mathrm{B}} \\ &= k \ln \left(\Omega_{\mathrm{A}} \Omega_{\mathrm{B}}\right). \end{aligned}$$

- All this demonstrates is that *Boltzmann’s relationship produces results which are consistent with what we expect; it is not a proof*.

Boltzmann's definition of entropy

- *At absolute zero we expect all of the particles to be in the ground state and so there is only one microstate, $\Omega = 1$*. Using the Boltzmann expression we therefore have that $\boldsymbol{S_0 = k \ln 1 = 0}$. The idea that all substances have zero entropy at absolute zero was introduced in year one.
- The constant $\boldsymbol{k}$ is, of course, what we call Boltzmann's constant. In SI it has the value 1.381×10^{-23} J K^{-1}. This constant is related to the gas constant, $\boldsymbol{R}$, by $\boldsymbol{R = N_A k}$, where $\boldsymbol{N_A}$ is Avogadro's constant.

Irreversible processes again

- Previously, we saw how the principle that all microstates are equally likely can be used to explain the occurrence of irreversible processes, and it was noted that for any finite change the probability of the reverse of such a process taking place was exceedingly small. Using Boltzmann's definition of entropy, a proper estimate of this probability can be made.
- Consider a change from an initial state with Ω_{init} microstates to a final state with Ω_{final} microstates. By Postulate 1, the probability of finding the system in any one of the microstates is the same, so the probability, P_{init}, of finding the system in any one of the microstates corresponding to the initial state is proportional to Ω_{init}, $P_{init} \propto \Omega_{init}$, and likewise the probability, P_{final}, of finding the system in any one of the microstates corresponding to the final state is proportional to Ω_{final}, $P_{final} \propto \Omega_{final}$.

Irreversible processes again

- Therefore the ratio of the time that the system will spend in the initial state to that spent in the final state is given by ($\boldsymbol{P_{init} / P_{final}}$). Using Boltzmann's definition of entropy, $\boldsymbol{S = k \ln \Omega}$, $\boldsymbol{\Omega}$ can be written as $\boldsymbol{\Omega = \exp(S / k)}$ and hence

$$\frac{P_{\text{init}}}{P_{\text{final}}} = \frac{\Omega_{\text{init}}}{\Omega_{\text{final}}} = \frac{\exp(S_{\text{init}}/k)}{\exp(S_{\text{final}}/k)} = \exp(-\Delta S/k)$$

where $\boldsymbol{\Delta S = (S_{final} - S_{init})}$.

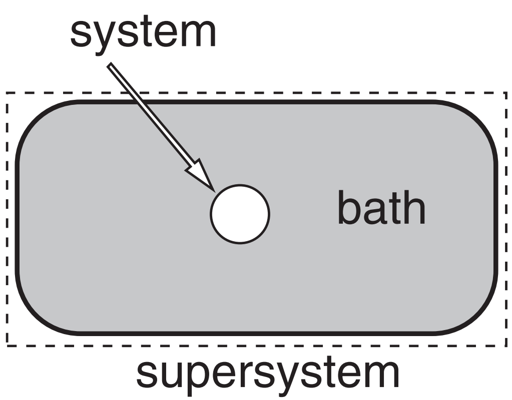
- For a finite change (such as a gas expansion), $\boldsymbol{\Delta S}$ is of the order of $\boldsymbol{nR}$, where $\boldsymbol{R}$ is the gas constant and $\boldsymbol{n}$ is the number of moles. Recalling that $k = R / N_A$, the ratio ($\boldsymbol{P_{init} / P_{final}}$) is therefore of the order of $\boldsymbol{\exp(-n N_A)}$, which is an incredibly small number. The system essentially spends no time in microstates corresponding to the initial state.

Example

- 1 mole of an ideal gas doubles in volume at constant temperature. Use classical thermodynamics to compute the entropy change. Estimate the probability of the reverse taking place.

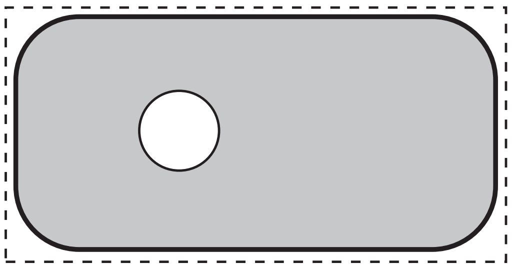
Canonical distribution

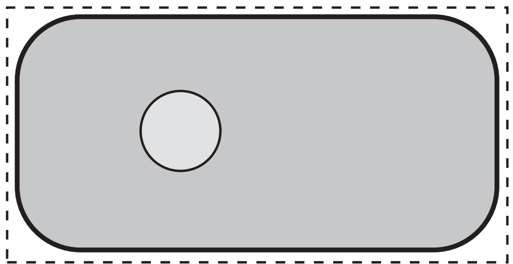
- Previous slides are concerned with an isolated system which is not what we normally have in chemical problems. More useful for us is a system which is in thermal equilibrium with its surroundings (i.e. a reaction taking place in a *thermostat*); in such a setup, heat is allowed to flow between the system and the surroundings. We now need to adapt the ideas above to deal with this situation.
- The arrangement we will consider is a system in contact with a large heat bath at temperature T; the system and the bath interact weakly, but sufficiently to maintain equilibrium. *Taken together the system and the bath form a supersystem, which we can regard as being isolated.* We seek an expression for the probability, $\boldsymbol{P_m}$, that the system is in a particular microstate, $\boldsymbol{m}$, with energy $\boldsymbol{E_m}$.



Calculation of P_m

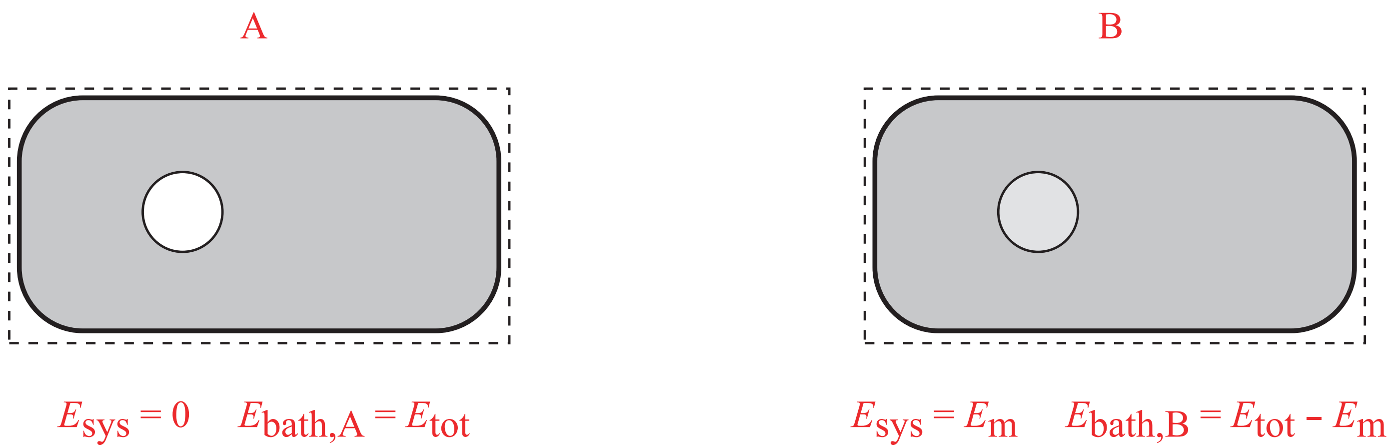
- The energy of the system is $\boldsymbol{E_{sys}}$, of the bath is $\boldsymbol{E_{bath}}$ and of the supersystem is $\boldsymbol{E_{tot}}$. Therefore, as the supersystem is isolated, $\boldsymbol{E_{tot} = E_{sys} + E_{bath}}$.
  The number of microstates of the system, the bath and the supersystem are $\boldsymbol{\Omega_{sys}}$, $\boldsymbol{\Omega_{bath}}$ and $\boldsymbol{\Omega_{tot}}$, respectively; therefore $\boldsymbol{\Omega_{tot} = \Omega_{sys} \times \Omega_{bath}}$.
- Compare the two situations shown below, in which the system is forced to be in a particular microstate. In **A** the energy of the system is zero, so all of the energy is in the bath: $\boldsymbol{E_{bath,A} = E_{tot}}$; in **B** the energy of the system is $\boldsymbol{E_m}$ so the energy of the bath is $\boldsymbol{E_{bath,B} = E_{tot} - E_m}$.





Calculation of P_m

- Since we are forcing the system to be in one microstate, $\Omega_{sys} = 1$ in both cases. However, as the energy of the bath is greater in **A** than in **B**, it follows that
  $\Omega_{bath,A} > \Omega_{bath,B}$
  and so $\Omega_{tot,A} > \Omega_{tot,B}$.
- Now we recall postulate 1 which says that all microstates of an isolated system are equally probable; in the present case our supersystem is isolated, so the postulate applies. It therefore follows that, as A has more microstates than **B**, **A** is a more probable arrangement than **B**.



Calculation of P_m

- What is happening here is that, as the energy of the system increases, it does so at the expense of the bath, thus decreasing the energy of the bath. In turn, this means that the number of microstates of the bath will decrease, and so the overall arrangement is less probable. Our task is now to turn this qualitative discussion into an exact expression for $\boldsymbol{P_m}$. What this boils down to is finding how $\boldsymbol{\Omega_{bath}}$ varies with $\boldsymbol{E_{bath}}$.
- The 'trick', such that it is, is to refer to an equation we derived earlier, which tells us how entropy varies with internal energy at constant volume:

$$\left(\frac{\partial S}{\partial U}\right)_V = \frac{1}{T}.$$

Calculation of P_m

- We are going to apply this to the bath, noting that in this case the internal energy $\boldsymbol{U}$ is the same at the $\boldsymbol{E_{bath}}$. Note also that, for an isolated system $\boldsymbol{S = k \ln \Omega}$, so here $\boldsymbol{S_{bath} = k \ln \Omega_{bath}}$. It therefore follows that

$$\frac{\mathrm{d} \ln \Omega_{\mathrm{bath}}}{\mathrm{d}E_{\mathrm{bath}}} = \frac{1}{kT}$$

  where we have taken constant volume to be implied.

- As the bath is very much larger than the system, we can use the change on going from $\boldsymbol{A}$ to $\boldsymbol{B}$ to approximate the derivative:

$$\frac{\ln \Omega_{\mathrm{bath,B}} - \ln \Omega_{\mathrm{bath,A}}}{E_{\mathrm{bath,B}} - E_{\mathrm{bath,A}}} = \frac{1}{kT}.$$

Calculation of P_m

- From what we have defined above $E_{bath,B} - E_{bath,A} = -E_m$, so the previous equation rearranges to

$$\ln \Omega_{\mathrm{bath,B}} - \ln \Omega_{\mathrm{bath,A}} = \frac{-E_{\mathrm{m}}}{kT}.$$

Changing to exponentials gives:

$$\Omega_{\mathrm{bath,B}} = \Omega_{\mathrm{bath,A}} \exp\left(\frac{-E_{\mathrm{m}}}{kT}\right)$$

- Finally we note that $\Omega_{bath,A}$ is not a function of E_m since we assumed that $E_{sys} = 0$ for arrangement ***A***. It therefore follows that

$$\Omega_{\mathrm{bath,B}} \propto \exp\left(\frac{-E_{\mathrm{m}}}{kT}\right).$$

- Now we recall that the probability of a particular arrangement is proportional to the number of microstates, so the probability of arrangement ***B*** is proportional to $\Omega_{bath,B}$; if we call this probability P_m (as the energy is E_m) we have

$$P_m \propto \exp\left(\frac{-E_m}{kT}\right)$$

Calculation of P_m

- We can work out the constant of proportion, ***B***, in the previous equation in the following way

$$P_m = B \exp\left(\frac{-E_m}{kT}\right)$$

- The total probability of occupying all the possible microstates of the system must be 1, so

$$\sum_m P_m = 1$$

where the sum over m is over all the microstates of the system. Computing this sum for both sides of the equation gives:

$$\begin{aligned}
\sum_m P_m &= B \sum_m \exp\left(\frac{-E_m}{kT}\right) \\
1 &= B \sum_m \exp\left(\frac{-E_m}{kT}\right) \\
\text{hence } B &= \frac{1}{\sum_m \exp\left(\frac{-E_m}{kT}\right)}.
\end{aligned}$$

Calculation of P_m

- Following on from this it is usual to define the partition function of the system, $\boldsymbol{Q_N}$, as

$$\boxed{Q_N = \sum_m \exp\left(\frac{-E_m}{kT}\right)}.$$

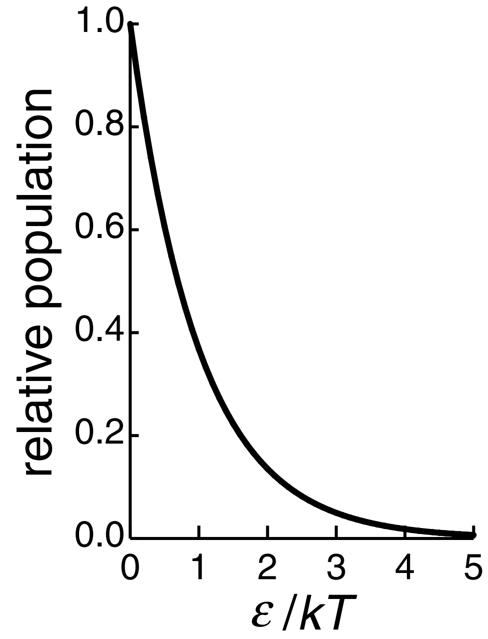
- It therefore follows that $\boldsymbol{B = 1/Q_N}$, and so the probability, $\boldsymbol{P_m}$, can be written

$$\boxed{P_m = \frac{1}{Q_N} \exp\left(\frac{-E_m}{kT}\right).}$$

- *This equation is called the canonical probability distribution function; it is the central result in statistical thermodynamics*. We will see later that the partition function is the key to calculating thermodynamic quantities.

Calculation of P_m

- This distribution function has a form which is very similar to the Boltzmann distribution. It says that the probability of a microstate drops exponentially as its energy increases; the rate of fall off depends exponentially on the dimensionless ratio E_m/kT. The graph given at the beginning (on the right) could just as well be a plot of P_m as of populations.
- *It is important to remember that P_m is the probability of a particular microstate of the system, not a population of an energy level.* A microstate of the system can be thought of as a particular arrangement of the particles in the energy levels. However, if the particles are interacting (due to appreciable intermolecular forces, for example), the microstate cannot be described in these terms, as the energy depends on the arrangement of the molecules. Nevertheless, the equation above still predicts the probability in terms of the partition function and the energy.



The canonical probability distribution function applies to a system in equilibrium with a heat bath. P_m is the probability that the system will be found in microstate m which has energy E_m.